Embedded System

by PUL079BEI008
by PUL079BEI014

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Notes

- Our subject code is CT 655.
- This document encorporates for both, electronics and computer department..
- If a question was asked in Regular Exam, it is kept in **boldened** form. If it was asked in back exam, it is in regular font.
- Months are marked as:
  - Ba: Baisakh
  - Jth: Jestha
  - Asa: Ashar
  - Shr: Shrawan
  - Bh: Bhadra
  - Ash: Ashwin
  - Ka: Kartik
  - Mng: Mangsir
  - Po: Poush
  - Ma: Magh
  - Ch: Chaitra

Contents

1 Introduction

(3 Hours / 4 Marks)

1.1 Embedded Systems Overview

1. What is Embedded system? [1] (**81 Ch**,**76 Bh**,**75 Bh**,**74 Bh**,**73 Bh** 73 Ma, 72 Ma, 70 Ma)
2. List out types of embedded system. Define them. [3] (**73 Bh**)
3. What are the features of an embedded system? Explain them with examples. [4] (**79 Ch**)
4. Differentiate embedded system and non-embedded system with suitable example. [3] (**75 Bh**)
5. What are the typical characterstics of embedded system ? [3] (**72 Ash**, 73 Ma)
6. How does digital camera satisfy embedded system characterstics. [1] (**72 Ash**)
7. How embedded system differing than general purpose computing system? [4] (71 Ma)
8. Clarify the statement 'Digital Camera is a good example of an Embedded System'. [3] (**74 Bh**)

1.2 Classification of Embedded Systems

1. Classify the embedded system based on generations. [4] (**76 Bh**, 72 Ma)

1.3 Hardware and Software in a system

1.4 Purpose and Application of Embedded Systems

1. What is a design metric. [1] (76 Ba, 75 Ba)
2. Explain any three design metrics of Embedded system. [3] (**81 Ch**)
3. Why power is an important design metrics in embedded system? [1] (**80 Ch**)
4. Define design metric. [1] (**78 Ch**)
5. Explain the various design metrics of embedded system. [3] (**80 Ch**,**78 Ch**)
   |→... all important design metrics —. [3] (76 Ba)
6. Explain different purpose of embedded system with examples. [4] (**77 Ch**,**71 Bh**) [3] (75 Ba)
7. Describe various application of embedded system. [3] (70 Ma)

2 Single Purpose Processor Design Issues

(4 Hours / 8 Marks)

2.1 Combination Logic

2.2 Sequential Logic

2.3 Custom Single-Purpose Processor Design

1. Develop algorithm; draw the state diagram and design the datapath of a custom single purpose processor that determines the greatest common divisor(GCD) of two numbers. Propose the block diagram of it's controller also. [8] (**81 Ch**)
2. Design a custom single purpose processor to find x^n showing all the steps involved. [8] (**80 Ch**)
   |→ Also, start the design from the function computing the desired result, FSMD, datapath and controller. [8] (**74 Bh**)
3. Design a dual purposed processor that takes in 10 numbers one at a time and finds the average and the maximum of the 10 numbers. Show algorithm, FSMD, FSM and Datapath for the processor. [8] (**79 Ch**)
4. Design a single purpose processor that gives the LCM of two digit 8-bit numbers. Start with the algorithm, translate it into state diagram, and state transition table. Also draw the required data path. [8] (**78 Ch**,**72 Ash**)
5. Design a single purpose processor that output Fibonacci numbers upto 'n' places. Start design from function computing the desired result, FSDM and sketch a probable datapath. [5] (**77 Ch**) [8] (**72 Ash**, 75 Ba)
6. Design a custom single purpose processor that display 1 or 0 if the input integer is prime or not. Show all the steps involved. [8] (**76 Bh**, 71 Ma)
7. Design a single purpose processor to determine the sum of digits of an integer. Start design from function computing the desired result, FSDM, datapath and controller. [8] (76 Ba)
8. Design a single purpose processor that calculates Factorial of an integer number 'n'. Start design from function computing the desired result, FSDM, datapath and controller. [8] (73 Ma)
9. Design a single purpose processor that determine the largest of four numbers. Start design from function computing the desired result, FSDM, datapath and controller. [8] (**73 Bh**)
   |→ Develop algorithm; draw the state diagram and design the datapath of a custom single purpose processor that determines the largest of four integers. Propose the block diagram of its controller also. [8] (**73 Bh**)
10. Design a dual purposed processor that takes in 5 numbers one at a time and finds the median and variance. Show algorithm, FSMD, FSM(controller) and Datapath for the processor. [8] (**70 Ma**)
11. Show the algorithm, FSMD, FSM, data-path and controller design for a dual-purpose processor that can calculate the factorial of a number and also check if that number is a prime number. [8] (71 Ma)

2.4 Optimizing Custom Single-Purpose Processors

1. What is optimization? [4] (**75 Bh**,**70 Bh**)
2. What are the parameter you consider for optimization of single purpose processor? [4] (**75 Bh**)
3. What are the optimization oppurtunities in single purpose processor. [3] (**77 Ch**)
4. Explain in detail about the optimization of custom single purpose processor with a suitable example. [6] (**70 Bh**) [8] (**71 Bh**, 72 Ma)

3 General Purpose Processor Design Issues

(6 Hours / 8 Marks)

3.1 Basic Architecture

1. Draw the basic architecture of General purpose processor and describe. [2] (**81 Ch**) [4] (**77 Ch**)
2. Define datapath and controller of a general purpose processor. [4] (**75 Bh**)
3. Difference between datapath and control unit. [2] (**71 Bh**)

3.2 Operation

1. Describe the operation of GPP interms of datapath and controller. [5] (**70 Bh**)
2. Explain the sub-operation of control unit with necessary illustration. [6] (**81 Ch**,**71 Bh**) [8] (71 Ma)
3. What do you mean by pipeline? [1] (**78 Ch**, 75 Ba, 72 Ma)
4. Explain with diagram about pipeling. [4] (76 Ba)
5. How can pipeline be implemented to improve performance of a processor? [4] (**79 Ch**)
6. Explain datapath operation and its instruction cycles. [5] (**72 Ash**, 72 Ma)
7. Explain 5 stage pipelining. [3] (**78 Ch**)
8. Explain 6 stage pipelining. [3] (**75 Ba**)

3.3 Programmer's View

1. Define Addressing modes. [1] (72 Mg)
2. Describe programmer's view in the context of software development for general purpose processor. [5] (**80 Ch**) [4] (**76 Bh**,**74 Bh**, 73 Ma)

3.4 Development Environment

1. Define Debugger, Simulator, testing and Emulator [4] (**80 Ch**)
   |→Explain testing and debugger. [3] (**70 Bh**)
2. Explain the design flow of embedded software development. [4] (**76 Bh**,**74 Bh**, 73 Ma)

3.5 Application-Specific Instruction-Set Processors

1. Write down/describe the features of a DSP. [4] (**79 Ch**,**78 Ch**,**73 Bh**)
   |→Explain DSP with characteristics and advantages. [4] (75 Ba)
2. Explain ASIC with its types. [4] (**75 Bh**)
3. Differentiate between application specific instruction set-processor and general purpose processor. Also discuss on issues related to selection of a particular processor. [8] (70 Ma)
4. How GPP differ from ASIPs? [4] (**73 Bh**,**70 Mg**)

3.6 Selecting a Microprocessor

1. What are the criteria to select the processor? [4] (**77 Ch**,**72 Ash**,**70 Mg**)
   |→Briefly explain suitable criteria for selecting Microprocessor in Embedded System. [4] (75 Ba)

3.7 General-Purpose Processor Design

4 Memory

(5 Hours / 8 Marks)

4.1 Memory Write Ability and Storage Permanence

1. Explain storage permanence of memory. [1.5] (**75 Bh**,**74 Bh**,**72 Ash**) [2.5] (**77 Ch**,**75 Bh**) [3] (**80 Ch**)
2. Explain the operation of storing data in one Time Programmable ROM. [3] (**78 Ch**)
3. Explain the memory write ability with suitable example. [1.5] (**75 Bh**,**74 Bh**,**72 Ash**) [2.5] (**77 Ch**,**76 Bh**)
4. Explain the operation of storing data in One Time Programmable ROM. [2] (75 Ba)
5. Why can't One Time Programmable ROM be reprogrammed? [2] (75 Ba)

4.2 Types of Memory

1. Show the differences between SRAM and DRAM. [3] (72 Ma)
2. What is enhanced DRAM built around the conventional DRAM core? [5] (71 Ma)
3. Describe ROM and introduce its types in detail. [6] (**70 Bh**)

4.3 Composing Memory

1. Compose 4Kx32 ROM using 2KX16 ROMs. [4] (**81 Ch**)
2. Comose 4Kx5 ROM using 1Kx2 ROM. [4] (**79 Ch**)
3. Design the internal architecture 4x4 ROM. [3] (**77 Ch**)
4. Compose a memory of size 2^{k+1} x 2n using 2^k x n sized memory. [5] (**78 Ch**)
   |→Construct 2^{k+1} x n and 2^k x 4n memories using 2^k x n memory modules. [4+4] (**73 Bh**)
   |→Compose 2^{k+1} x m memory using 2^k x m memories. [3] (70 Ma)
5. Compose 4Kx16 ROM using 1Kx8 ROMs. [5] (76 Ba)
6. Compose 1Kx8 ROMs into a 4Kx8 ROM. [4] (75 Ba)
   |→Compose 1Kx8 ROMs into a 2Kx16 ROM. [5] (72 Ma)
7. Sketch the internal design of a 4 x 3 ROM. [2] (**70 Bh**)
8. Construct a ROM/PROM to store data shown in table below: [5] (**80 Ch**)

Address		Information			
X	Y	F1	F2	F3	F4
0	0	0	1	0	0
0	1	0	1	1	1
1	0	1	0	1	1
1	1	1	0	0	1

9. Design a ROM to store the following information. [5] (**75 Bh**)

X	Y	Z	F1	F2	F3	F4
0	0	0	0	0	1	0
0	0	1	1	1	0	0
0	1	0	0	1	0	1
0	1	1	1	1	1	1
1	0	0	0	0	1	1
1	0	1	0	1	0	1
1	1	0	1	0	1	0
1	1	1	0	0	1	1

10. Design a ROM to store the following information. [5] (**75 Bh**)

X	Y	Z	F1	F2
0	0	0	1	0
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	0	0
1	0	1	1	1
1	1	0	0	1
1	1	1	1	0

4.4 Memory Hierarchy and Cache

1. What is cache memory? [2] (**71 Bh**)

2. Explain fully associative cache mapping technique. [4] (**81 Ch**)

3. Explain the impact of cache size in performance of cache memory. [4] (**79 Ch**)

4. Explain in brief about programmer's view for general purpose processor. [4] (73 Ma)

5. Write cache mapping techniques. [6] (**71 Bh**)

6. Explain associative cache mapping technique. [5] (**72 Ash**)
   |→with its merits and demerits. [5] (**74 Bh**)

7. Explain replacement algorithms used in cache memory. [3] (**76 Bh**, 71 Ma)
   |→What are the basic techniques for cache mapping? [4] (73 Ma)

8. Determine which cache memory is better. [3] (76 Ba)

	Size	Miss Rate	Hit Cost	Miss Cost
Cache A	2 KB	15%	2 cycles	30 cycles
Cache B	8 KB	05%	4 cycles	30 cycles

5 Interfacing

(6 Hours / 8 Marks)

5.1 Communication Basics

1. Define interfacing. [1] (**80 Ch**,**77 Ch**)
2. Explain the needs of interfacing. [2] (**80 Ch**,**77 Ch**)
3. Explain control methods used for communication in interfacing. [3] (**78 Ch**)

5.2 Microprocessor Intefacing: I/O Addressing, Interrupts, DMA

1. What is the difference between memory-mapped I/O and standard I/O. [3] (**72 Ash**)
2. What is interrupt? [2] (**76 Bh**,**75 Bh**)
3. How interrupt needed in digital devices? [3] (**76 Bh**)
4. How does Interrupt driven I/O operates with Fixed ISR location. Clearly explain the steps by taking suitable example. [8] (**79 Ch**)
5. Explain summary of flow of actions of interrupt driven I/O using fixed ISR location. [2] (**75 Bh**)
6. Describe the major steps to be followed in interrupt processing by a processor. [5] (72 Ma)
7. Explain the operation of peripheral to memory transfer without DMA, using vector interrupt. [5] (**72 Ash**) [8] (71 Ma)
8. Describe the purpose of the DMA controller. [2] (70 Ma)
9. Draw the flow of actions between peripheral and memory using DMA. [2] (70 Ma)
10. Design an interface circuit of a microprocessor with 16-bit address 2 RAMs and 2 ROMs of 8 KByte each. [8] (73 Ma)
11. What are the differences between strobe and handshake protocol? [4] (**71 Bh**)

5.3 Arbitration

1. What is arbitration? [1] (**75 Bh**) [2] (**73 Bh**)
2. Explain different types of arbitration methods used in peripherals devices to gain control of system bus. [4] (**70 Bh**)
3. Explain Priority arbitration. [3] (**73 Bh**) [5] (**80 Ch**)
   |→with proper illustration and types. [5] (**77 Ch**)
   |→and compare with daisy-chain arbitration. [5] (**74 Ch**)
4. Explain Daisy-chain arbitration. [3] (**73 Bh**) [4] (**71 Bh**)
   |→with neat diagram. [3] (**75 Bh**)
5. Describe Daisy-Chain arbitration with the help of a block diagram and steps. [5] (**78 Ch**)
   |→Explain structure, operation, advantages and disadvantages of daisy-chain arbitration method. [6] (76 Ba)
6. Write a sumamry of flow of actions for interrupt driven I/O using fixed ISR location. [3] (**76 Bh**)

5.4 Multilevel Bus Architecture

1. Describe two-level bus architecture in detail. [3] (**74 Bh**)

5.5 Advanced Communication Principles

1. Describe the advanced communication principles used in embedded systems. [4] (70 Ma)

2. Describe the significance of I^2C serial communication protocol. [4] (**70 Ch**)

6 Real-Time Operating System (RTOS)

(8 Hours / 12 Marks)

6.1 Operating System Basics

1. How RTOS is different from GPOS? [3] (**81 Ch**,**79 Ch**) [4] (**71 Bh**, 73 Ma)
2. Differentiate between multiprocessing and multi tasking in RTOS. [3] (**72 Ash**)
3. Define kernel. [1] (**73 Bh**, 76 Ba)
4. Explain importance of kernel operating system in embedded systems. [4] (71 Ma)
5. Explain types of kernel. [2] (76 Ba)
6. Briefly describe the kernel operating system services. [4] (**72 Ash**)
7. Differentiate between monolithic kernel and micro kernel. [2] (**73 Bh**)
8. Explain file system handling of kernel. [3] (76 Ba)
9. Describe basic fucntion of real time kernel. [5] (**77 Ch**)

6.2 Task, Process, and Threads

1. Differentiate between process and thread. [3] (**81 Ch**,**79 Ch**, 70 Ma) [4] (73 Ma)
   |→any 4. [2] (**74 Bh**)
   |→atleast 6. [3] (**71 Bh**)
2. Compare task, process and threads. [4] (**77 Ch**)
3. Define threads. [1] (75 Ba)
   |→in RTOS. [3] (72 Ma)
4. Differentiate between user level thread and kernel level thread. [2] (75 Ba) [3] (**70 Bh**)
5. Briefly explain the different states of task. [4] (**78 Ch**)
6. Explain process life cycle with process state diagram. [3] (**76 Bh**)
7. Write a simple multithreading program in C. [3] (70 Ma)

6.3 Multiprocessing and Multitasking

1. Explain context switching. [3] (**70 Bh**)
   |→with suitable diagram. [4] (**74 Bh**)
2. What are the different situations when context switching is necessary? [2] (**80 Ch**)
3. Explain the details of steps involved in context switching. [4] (**80 Ch**)

6.4 Task Scheduling

1. Define task scheduling, list out its types and explain the factors affecting on selection of scheduling algorithm. [4] (75 Ba)

2. Three processes P1, P2 and P3 with estimated completion time 5, 8, 7 ms respectively enters the ready queue together. Calculate WT, TAT for each process and calculate AWT and ATAT using Round Robin Pre-emptive scheduling algorithms with time slice of 2 ms. [6] (**75 Bh**)

3. Consider three processes with process IDs P1, P2, P3 with estimated completion time 9, 6, 3 ms respectively, enters the ready queue together in order P1, P2, P3. Calculate WT and TAT for each process and average waiting time and average turn around time in RR (Round-Robin) algorithm with time slice 2 ms. Assume there is no I/O waiting for the process. [4] (73 Ma)
   |→slice = 3 ms. [5] (72 Ma)
   |→completion times = 6, 4, 2. [6] (**72 Ash**)

4. Three processes with process IDs P1, P2, P3 with estimated completion times 4, 6, 5 milliseconds and priorities 1, 0, 3 respectively enters the ready queue together. Calculate WT and TAT for each process andd average WT and TAT in preemptive priority based scheduling algorithm, by assuming zero is the highest priority and three is the lowest priority. [5] (**71 Bh**)

5. Three process P1, P2, and P3 with estimated completion time 4, 10, 5 ms and priorities 1, 3, 2 respectively enter the ready queue together. A new process P4 with estimated completion time 3ms and priority 0 enters the ready queue after 5ms of start of operation. Calculate WT, TAT for each process and calculate AWT and ATAT using preemptive priority-based scheduling algorithms. [6] (**81 Ch**,**74 Bh**)

6. Three processes with process IDs P1, P2, P3 with estimated completion times 6, 8, 2 milliseconds respectively enters the ready queue together. Process P4 with estimated execution completion time 4 milliseconds enters the ready queue after 1 millisecond. Calculate the waiting time and turn around time for each process and the AWT and TAT in the non-preemptive shortest job first scheduling. [3] (**70 Bh**) [5] (**77 Ch**, 75 Ba)

7. Three processes with process IDs P1, P2, P3 with estimated completion time 8, 6, 10 ms and priorities 0, 3, 2 (0-highest, 3-lowest) respectively, enters the ready queue together in order P1, P2, P3 (assume only P1 is present in the Ready queue when the scheduler picks it up and P2 and P3 enter ready queue after that). Now, a process P4 with estimated completion time 6 ms and priority 1 enters the ready queue after 5 ms of execution of P1. Calculate waiting time and TAT for each process and average waiting time and TAT. Assume there is no I/O waiting for the processes and priority-based scheduling. [5] (**76 Bh**)
   |→estimated completion times: 7, 8, 5, 10. [6] (76 Ba)
   |→estimated completion times: 10, 5, 7, 6. Priorities: 1, 3, 2, 0. P4 enters ready queue after 5 ms. [5] (**73 Bh**) [8] (71 Ma)

8. Consider Four processes with process IDs P1, P2, P3 and P4 with estimated completion time 7, 4, 12, 3 ms, enters the ready queue together. Process P_5 with completion time 4 ms enters the ready queue after 2 ms of the scheduling of process P_1 starts. Calculate waiting time and turnaround time for each process and average waiting times and average turnaround time in Preemptive Shortet Job First Algorithm. Assume there is no I/O waiting for the process. [6] (**80 Ch**)

9. Four processes with process IDs P1, P2, P3 and P4 with priorities 0, 2, 4, 5 and estimated completion time 5, 4, 6 and 4 ms respectively enter the ready queue together. Two new processes P5 and P6 with priorities 1, 3 and estimated completion time 2 and 3 ms respectively enters the ready queue:

P5 enters 1 ms of start of execution while P6 enters ready queue after 3 ms of start of execution of P4. Calculate WT, TAT, AWT, ATAT assuming that there is no I/O waiting for the process using pre-emptive priority based and SJF/SRT algorithm. [6] (**79 Ch**)

10. Consider four processes with process IDs P1, P2, P3 and P4 with estimated completion time 53, 17, 68, 24 ms respectively, enters the ready queue together in order P1, P2, P3 and P4. Calculate waiting time and turn around time for each process and average waiting times and average turn around time in Round Robin algorithm with time slice 4 ms. Assume there is no I/O waiting for the process. [6] (**78 Ch**)

6.5 Task Synchronization

1. What is task synchronization [1] (**76 Bh**, 72 Ma)
2. Explain the task synchronization using 'Busy/Wait'. [3] (**76 Bh**, 72 Ma)
3. Describe mutual exclusion through sleep and wake for task synchronization. [4] (**75 Bh**)
4. Explain the conditions favoring deadlock situation. [2] (**75 Bh**)
5. Define deadlock. [2] (**73 Bh**)
6. explain the Coffman's condition for deadlock. [2] (**73 Bh**) [3] (**70 Bh**)
   |→List Coffman Condition for deadlock. [2] (**78 Ch**)

6.6 Device Drivers

7 Control System

(3 Hours / 8 Marks)

7.1 Open-loop and Close-loop Control System Overview

1. Differentiate between open loop and closed loop control system. [2] (**78 Ch**, 76 Ba) [3] (**75 Bh**,**72 Ash**) [4] (**79 Ch**)

2. With neat diagram write the steps for designing closed loop control system. [5] (**72 Ash**)

3. Draw a clear block diagram of a closed-loop control system and explain its various parts. [4] (71 Ma)

4. Which system is better for control system? [2] (76 Ba)

5. What are the evaluation steps used in general control systems input? [3] (**71 Bh**)

6. Explain different performance metrics of control system using suitable performance response diagram. [4] (**77 Ch**)

7. Define the following terms used in control system: Controller, Plant, Actuator. [3] (**70 Bh**)

7.2 Control System and PID Controllers

1. What are the challenges of modeling a real physical system and how can you overcome it? [3] (**74 Bh**)

2. Differentiate between P, PI, PD, and PID controllers. [4] (71 Ma)

3. Explain controller design using PD and PI controller. [5] (**71 Bh**)

4. Explain the metrics used to measure the control objectives. [4] (**76 Bh**,**73 Bh**)

5. What is PID controller? [2] (**81 Ch**, 76 Ba)

6. Draw a typical block diagram of PID control system. [2] (**75 Bh**)

7. Explain the operation of a PID control with a block diagram. [5] (**70 Bh**)

8. Explain types of PID with real time application in embedded system. [2] (76 Ba)

9. Explain PID control system with block diagram and equations. [4] (**79 Ch**)

10. What is the purpose of PID tuning, and what are the benefits of computer based control implementations? [4] (70 Ma)

7.3 Software coding of PID Controller

1. Write an algorithm for a PID control. [4] (**77 Ch**)
   |→Write an algorithm to implement the PID controller in software. [4] (**73 Bh**)
   |→Write the pseudo-code for PID controller. [4] (70 Ma)

2. Explain software coding of PID controller. [4] (**76 Bh**)
   |→Write an algorithm to implement the PID controller in software. [5] (**74 Ch**)

7.4 PID Tuning

7.5 Design

1. Design control system for an automobile cruise control in open-loop control system using proportional control. [6] (**81 Ch**,**78 Ch**) [8] (72 Ma)
   |→Draw the block diagram of closed-loop control system for speed control of an automobile and explain the conditions for no unbound and no oscillation showing all the design steps. [8] (75 Ba, 73 Ma)

2. Draw the block diagram of a closed-loop control system to control speed in an automobile. [2] (**80 Ch**)

3. Design the control system showing all the steps and derive the condition for no oscillation and no unbound output. [6] (**80 Ch**)

4. Describe PID tuning. [5] (**75 Bh**)

8 IC Technology

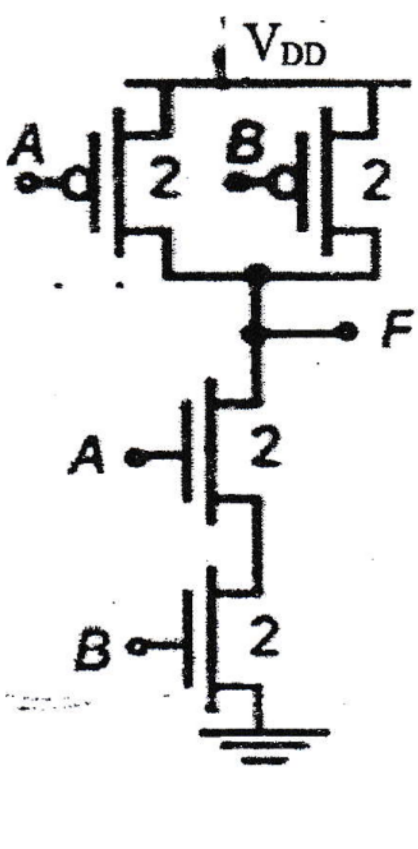
(3 Hours / 8 Marks)

8.1 Full-Custom VLSI IC Technology

1. What is Full Custom IC technology. [2] (**79 Ch**)
2. Describe briefly about Full custom VLSI technology. [4] (75 Ba)
3. Explain the steps involved in full-Custom IC design. [5] (**80 Ch**)
4. What are the merits and demerits of full custom IC technology? [2] (**79 Ch**,**76 Bh**) [3] (**72 Ash**) [4] (**78 Ch**)

8.2 Semi-Custom ASIC IC Technology

1. What is Semi-Custom IC technology? [2] (**79 Ch**,**77 Ch**)
2. Explain teh two broad categories of Semi-Custom IC technology. [3] (**74 Ch**, 73 Ma)
3. Briefly explain semi-custom (ASIC) IC technology and write about its types. [3] (76 Ba)
4. What are the merits and demerits of semicustom IC technology? [2] (**81 Ch**)
5. Explain the steps involved in IC manufacturing. [4] (**79 Ch**,**70 Bh**)
   |→with neat block diagram. [5] (70 Ma)
6. What are IC manufacturing steps in semiconductor industries? [3] (71 Ma)
7. Show the top-down view of the circuit F = xz + y on an IC. [4] (**70 Bh**)
8. Explain IC manufacturing steps of circuit given below. [5] (76 Ba)



8.3 Programming Logic Device (PLD) IC Technology

1. Explain gate-array semi-custom IC technlogy and standard cell semi-custom IC technology. [4] (**81 Ch**)

2. Draw the top-level diagram of a NAND gate using CMOS. [3] (**80 Ch**)

3. What is photolithography and what are its types? [4] (72 Ma)

4. Explain the importance of photolithography in IC manufacturing. [5] (**74 Bh**, 73 Ma)
   |→ with suitable example. [8] (**71 Bh**)

5. Describe the alignment and exposure techniques in photolithographic process in IC fabrication. [5] (71 Ma)

6. Explain the basic steps of photo lithography process. [4] (**78 Ch**,**77 Ch**, 72 Ma) [5] (**72 Ash**) [6] (**76 Bh**)
   |→ with appropriate figure. [4] (75 Ba)

7. Explain Full-Custom, Semi-Custom and PLD schemes used in IC technology with steps and comparision. [8] (**73 Bh**)

8. List the three major IC technologies with brief definitions. [3] (70 Ma)

9 Microcontrollers in Embedded Systems

(3 Hours / 8 Marks)

9.1 Intel 8051 microcontroller family, its architecture and instruction sets

1. How are microcontrollers different from microprocessors? [2] (**78 Ch**)
2. What are the important operational features of 8051 microcontroller? [2] (**80 Ch**,**78 Ch**) [3] (**73 Bh**) [4] (**81 Ch**)
3. Draw the pin diagram of 8051 microcontroller and explain ports 1 and 2 only. [3] (**72 Ash**)
4. Show the internal structure of the 8051 microcontroller. [4] (**70 Bh**)
5. Explain data memory organization in 8051 microcontroller. [4] (**79 Ch**)
6. Which I/O (port based or bus based) is used in 8051 Microcontroller, describe. [2] (76 Ba)
7. Explain port based and bus based I/O. [3] (72 Ma)
8. Explain about PORT 3 of 8051 microcontroller. [2] (76 Ba)
9. Explain the addressing modes used in 8051 microcontroller with example. [4] (75 Ba, 72 Ma, 71 Ma)
10. Provide a comparison chart of the 8051 family members. [4] (**70 Bh**)

9.2 Programming in Assembly Language

1. Write an assembly language programming to alternatively blink the 8 LEDs connected at port 2 of the 8051 microcontroller. [4] (**81 Ch**, 72 Ma)
2. Write an assembly program for 8051 to count number of 0's in an 8-bit data stored in ROM at 40H and result in RAM at 41H. [4] (**79 Ch**)
3. Assume that XTAL = 11.0592 MHz, write a program to generate a square wave of 1 kHz frequency on pin P1.5 of 8051 microcontroller using timer. [2+4] (**78 Ch**)
4. Generate a periodic square wave having a period of 15 ms and a duty cycle of 20% in 8051 using assembly programming. The waveform should be produced at pin zero of port two (P2.0). The XTAL frequency is 11.0592 MHz and use Timer 1 in mode 0 (13-bit timer mode). [8] (**76 Bh**)
5. Write an assembly program to design a down counter that counts from 99 to 00. [6] (**77 Ch**)
6. Using 8051 instructions, control rate of blink of LED at pin P1.1 by two switches at P2.1 and P2.2 (One switch to increase rate of blink, another to decrease the rate of blink). [6] (76 Ba)
7. Write an assembly language programming to blink 8 LED connected at Port 2 of 8051 microcontroller. [4] (75 Ba)
8. Describe the different purpose of port 3 and port 2 of 8051 microcontroller. Write an assembly language programming for 8051 microcontroller to read the data from switches connected at port 1 and send it to port 2 for display in LED. [4+4] (73 Ma)

9. Write a program in assembly language that computes a precise 5-millisecond delay using 8051 microcontroller. [4] (71 Ma)

10. Write a program using C-programming language to find the sum between two 8-bit BCD data stored in RAM locations 50H and 51H and store the BCD sum at RAM locations 52H and 53H. [5] (**72 Ash**)

9.3 A simple interfacing example with 7 segment display

1. What is seven segment display and write its types. [2] (**75 Bh**)

2. Explain different configuration for seven segment display. [2] (**77 Ch**)

3. Write an Assembly language program to display 9 to 0 in common anode 7-segment display connected at port 1 and show interfacing diagram also. [6] (**80 Ch**)
   |→Write an assembly program to display 0 to 9 in seven segments display. [5] (**73 Bh**)

4. Design a circuit with 7 segments display which is used as a counter watch which display second and minute. [6] (**75 Bh**)

5. Write an assembly and C program to display 0 to 9 in seven segment display. [8] (**71 Bh**)

6. Write the code for BCD counter to display 0 to 9999 in seven segment using VHDL. [8] (**80 Bh**)

7. Write 8051 program and draw circuit diagram to display number from 99 to 00 in seven segment display. The program should be written in both assembly and C. [8] (70 Ma)

10 VHDL

(4 Hours / 8 Marks)

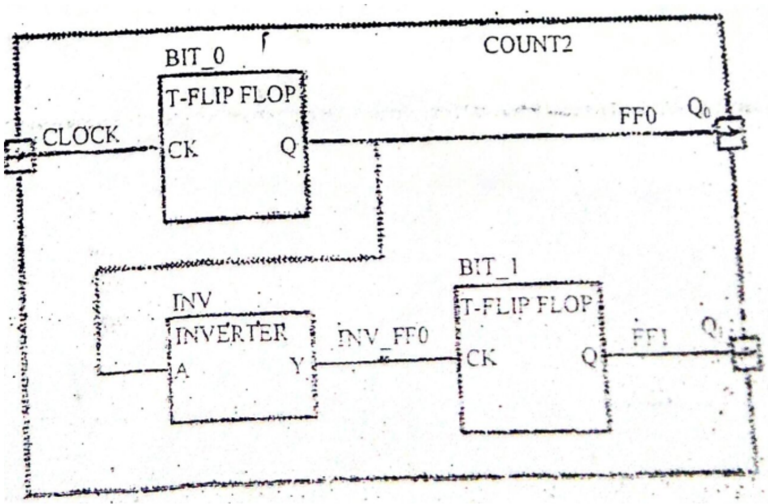
10.0.1 VHDL overview

1. How does a FPGA differ from a microcontroller? [1] (70 Ma)
2. What is structural style of architecture in VHDL? [3] (**81 Ch**)
3. Explain different models of VHDL. [3] (**75 Bh**)
4. Define behavioral model. [3] (**80 Ch**)
   |→Define behavioral modeling. [2] (72 Ma)
5. Explain different architecture model of VHDL. [2] (**79 Ch**)
6. What's the use of VHDL code in embedded system? [2] (76 Ba)
7. Explain COMPONENT with its declaration. [3] (75 Ba)
8. Explain PROCESS in VHDL. [3] (73 Ma)
9. How entity is declared in VHDL programming? [3] (71 Ma)

10.0.2 Finite state machine design with VHDL

1. Write a VHDL code for a JK flip-flop. [5] (**81 Ch**)
   |→using PROCESS. [5] (75 Ba)
2. Write a VHDL code for 4-bit BPA in structural model. [5] (**80 Ch**)
3. Write a VHDL code to design a traffic light controller unit with necessary assumptions using state machine. [6] (**79 Ch**)
4. Draw the state diagram for a sequence detector for the sequence 1011 and then develop a VHDL code based on the state diagram. [5] (**78 Ch**)
5. Using structural model, write a code in VHDL to implement full adder. [8] (**77 Ch**)
6. Write a VHDL code for a full adder using two half adders and one OR gate in structural model. [5] (**75 Bh**)
   |→Write a VHDL code for a full adder using 2 half adder as component. [5] (73 Ma)
7. Write an algorithm and VHDL code for custom single purpose processor that calculates the Greatest Common Divisor (GCD) of two numbers as Finite State Machine. [6] (**76 Bh**) [2+6] (**73 Bh**) [3+5] (**71 Bh**)
8. Design and write a code for Decoder using VHDL. [6] (76 Ba)
9. Write VHDL code for 2-input multiplexer. [6] (72 Ma)
10. Write an algorithm and VHDL code for a custom processor that calculates LCM of two numbers. [5] (**72 Ash**)

11. A circuit composed of two T-flip flops and an inverter shown in figure below, convert it to VHDL code. [5] (71 Ma)



12. Design a sequence detector for the string "1101", that outputs a one when the input matches this string, show the fSM and its VHDL implementation. [7] (70 Ma)

11 No idea where this is from

1. Using the simplified revenue model, derive the percentage revenue loss equation for any rise angle, rather than just 45 degrees. [2] (**70 Bh**)